Hersh Kumar

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Research Interests

My research focuses on theoretical and computational physics, specifically quantum simulation of lattice gauge theories. I work on methods for preparing and simulating scattering processes on quantum computers, such as digital quantum simulation and dissipative state preparation.

I am also interested in the application of classical computation to model many-body quantum systems, particularly the use of neural networks as ansatze for interacting quantum systems.

Education

University of Maryland, College Park, College Park, MD USA

Ph.D Student, Physics, Advisor: Prof. Zohreh Davoudi **2024 – present**

University of Maryland, College Park, College Park, MD USA

B.S in Physics with High Honors **2020 – 2024**
minor in Computer Science

High Honors thesis: *Neural Network Ansatze for Bosonic and Fermionic Quantum Systems in One Dimension*

Publications

Paulo F. Bedaque, **Hersh Kumar**, Suryansh Rajawat, Gregory Ridgway. *Neural Wavefunctions in Quantum Field Theory I: Asymptotic Freedom.* Preprint on arXiv. (2026)

Paulo F. Bedaque, Jacob Cigliano, **Hersh Kumar**, Srijit Paul, Suryansh Rajawat. *Trapped Fermions Through Kolmogorov-Arnold Wavefunctions.* Preprint on arXiv. (2025)

Paulo F. Bedaque, Jacob Cigliano, **Hersh Kumar**, Srijit Paul, Suryansh Rajawat. *Kolmogorov-Arnold Wavefunctions.* Phys. Rev. C 112, 054002 (2025)

Paulo F. Bedaque, **Hersh Kumar**, Andy Sheng. *A Machine Learning Approach to Trapped Many-Fermion Systems.* Phys. Rev. C 112 (1), 014002 (2025)

Paulo F. Bedaque, **Hersh Kumar**, Andy Sheng. *Neural Network Solutions of Bosonic Quantum Systems in One Dimension.* Phys. Rev. C 109, 034004 (2024)

Edison M. Murairi, Michael J. Cervia, **Hersh Kumar**, Paulo F. Bedaque, Andrei Alexandru. *How many quantum gates do gauge theories require?* Phys. Rev. D 106, 094504 (2022)

Code

Languages: Python, C++, LaTeX, HTML/CSS, Javascript, Java
Tools: Linux, `git`, Jupyter Notebook, Excel, Jax, MPI, OpenMP, CUDA
GitHub: github.com/hershkumar

Talks

Lattice 2026

Dissipative Preparation of Vacuum and Meson States in a 1+1D $\mathbb{Z}_2$ Lattice Gauge Theory **July 2026**

Gave an invited talk at Lattice 2026 on the applications of dissipative state preparation algorithms to lattice gauge theories on quantum computers, and the potential for performance guarantees for state preparation. Slides for the talk can be found here.

Introduction to Machine Learning

Neural Networks, Gradient Descent, and Backpropagation **June 12th, 2025**

Invited to give a pedagogical talk on neural networks, gradient descent, and backpropagation for the University of Maryland Particle Astrophysics group. Slides for the talk can be found here.

Undergraduate High Honors Presentation

Neural Network Ansatze for Quantum Systems in One Dimension **May 10th, 2024**

Delivered a senior thesis talk on the usage of neural networks as variational ansatze for bosonic and fermionic systems in one dimension. Slides for the talk can be found here, and my senior thesis can be found here.
Advisor: Prof. Paulo Bedaque

University of Maryland Physics Undergraduate Colloquium

Variational Monte Carlo with a Neural Network Ansatz **October 10th, 2023**

Delivered an informal talk on the applications of neural networks as ansatze for many-boson systems in one dimension. The talk was geared towards undergraduate physics students. The slides I created for the talk can be found here.

Teaching

Graduate Teaching Assistant: University of Maryland **Aug. 2024 – Dec. 2024**

Quantum Physics II (Fall 2024). Wrote and graded homeworks and exams, held office hours, and gave lectures.

Undergraduate Teaching Assistant: University of Maryland **2022 – 2023**

Quantum Physics II (Fall 2023), Introduction to Thermodynamics and Statistical Mechanics (Fall 2022, Spring 2023). Graded homeworks, held office hours, assisted with problem solving sections.